Linear Algebra and Its Applications

Section 4.9: Applications to Markov Chains

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

Contents

4.9 Applications to Markov chains 3

Practice problem 4.9.P1: A city resident's next two locations 3

Practice problem 4.9.P2: Test the steady-state equation rather than its appearance 3

Practice problem 4.9.P3: A long-run migration percentage 3

Exercise 4.9.1: Radio listeners at two station breaks 3

Exercise 4.9.2: Three equally treated food choices 4

Exercise 4.9.3: Healthy and ill students 4

Exercise 4.9.4: Good, indifferent, and bad weather 4

Exercise 4.9.5: Normalize a two-state balance equation 5

Exercise 4.9.6: A second two-state equilibrium 5

Exercise 4.9.7: Three-state balance with two useful equations 5

Exercise 4.9.8: A steady state despite zero transition entries 5

Exercise 4.9.9: One zero can disappear under squaring 5
Exercise 4.9.10: An absorbing state prevents regularity 6
Exercise 4.9.11: Long-run radio preferences 6
Exercise 4.9.12: No long-run preference between symmetric foods 6
Exercise 4.9.13: The long-run illness rate 6
Exercise 4.9.14: Long-run good weather 6
Exercise 4.9.15: A migration percentage in the long run 6
Exercise 4.9.16: Allocate a car fleet across three locations 7
Exercise 4.9.17: A stochastic matrix has eigenvalue one 7
Exercise 4.9.18: An explicit steady state for every two-state matrix 7
Exercise 4.9.19: A row of ones records total probability 7
Exercise 4.9.20: The square is stochastic 8
Exercise 4.9.21: Observe convergence of all columns of powers 8
Exercise 4.9.22: A reproducible comparison of two numerical methods 9

4.9 Applications to Markov chains

We use column probability vectors. Entry p_{ij} is the probability of moving *from state j to state i*, so each column of the transition matrix sums to 1 and its entries are nonnegative. The distribution evolves by $x_{k+1} = Px_k$. A steady-state vector is a probability vector q with $Pq = q$; solve $(P-I)q = 0$ together with $\sum_i q_i = 1$. A stochastic matrix is regular if some positive integer power has every entry strictly positive. For a regular matrix, Theorem 18 guarantees a unique steady-state vector and convergence $P^k x_0 \to q$ for every initial probability vector.

Practice problem 4.9.P1: A city resident's next two locations

The migration matrix from Example 1 is $M = \left(\begin{smallmatrix} .95 & .03 \\ .05 & .97 \end{smallmatrix}\right)$, with city first and suburbs second. Starting from $x_0 = (1, 0)$, $x_1 = Mx_0 = (.95, .05)$. Thus (a) the probability of living in the suburbs next year is 5%. A second multiplication gives $x_2 = (.95^2 + .03(.05), .05(.95) + .97(.05)) = (.904, .096)$. Thus (b) the probability after two years is 9.6%.

Practice problem 4.9.P2: Test the steady-state equation rather than its appearance

For the given P, q, direct multiplication gives

$$\begin{pmatrix} .6 & .2 \\ .4 & .8 \end{pmatrix} \begin{pmatrix} .3 \\ .7 \end{pmatrix} = \begin{pmatrix} .18 + .14 \\ .12 + .56 \end{pmatrix} = \begin{pmatrix} .32 \\ .68 \end{pmatrix} \neq q.$$

Although q is a probability vector, it is not a steady-state vector.

Practice problem 4.9.P3: A long-run migration percentage

Solve $Mq = q$: the first equation reduces to $.05q_1 = .03q_2$. Together with $q_1 + q_2 = 1$, this gives $q = (3/8, 5/8)$. Every entry of M is positive, so it is regular; Theorem 18 makes this the limiting distribution. Thus 62.5% will live in the suburbs after many years.

Exercise 4.9.1: Radio listeners at two station breaks

(a) Build columns from current stations.

Order the states as news, music for both rows (next station) and columns (current station). The probabilities give

$$P = \begin{pmatrix} .7 & .6 \\ .3 & .4 \end{pmatrix}.$$

Column one describes a news listener; column two describes a music listener. Each column sums to 1.

(b) Use the stated initial audience.

Everyone initially listens to news, so $x_0 = (1, 0)$.

(c) Count the breaks, not the elapsed minutes.

At 9:25 there have been two breaks, at 8:30 and 9:00. Then $x_1 = (.7, .3)$ and $x_2 = (.7(.7) + .6(.3), .3(.7) + .4(.3)) = (.67, .33)$. Thus 33% will be listening to music.

Exercise 4.9.2: Three equally treated food choices

(a) Use foods 1, 2, 3 in both state lists.

The probability of retaining the current food is .6; the other two possibilities each have probability .2. Thus

$$P = \begin{pmatrix} .6 & .2 & .2 \\ .2 & .6 & .2 \\ .2 & .2 & .6 \end{pmatrix}.$$

(b) Apply two transitions after the initial trial.

Starting at food 1 gives $x_0 = (1, 0, 0)$ and $x_1 = (.6, .2, .2)$. The second component of x_2 is $.2(.6) + .6(.2) + .2(.2) = .12 + .12 + .04 = .28$. The requested probability is 28%; a single multiplication would answer the first trial after the initial trial, not the second.

Exercise 4.9.3: Healthy and ill students

(a) Define the state order and fill the columns.

Use healthy, ill. A healthy student becomes ill with probability .05; an ill student becomes healthy with probability $1 - .55 = .45$. Therefore

$$P = \begin{pmatrix} .95 & .45 \\ .05 & .55 \end{pmatrix}.$$

(b) Advance Monday's probability vector twice.

Monday's $x_0 = (.8, .2)$ gives Tuesday $x_1 = (.95(.8) + .45(.2), .05(.8) + .55(.2)) = (.85, .15)$. Wednesday gives $x_2 = (.875, .125)$. The ill percentages are 15% and 12.5%.

(c) Use a different initial condition for a currently healthy student.

From $(1, 0)$, after one day the vector is $(.95, .05)$, and after two days the healthy probability is $.95^2 + .45(.05) = .925$. Thus the requested probability is 92.5%.

Exercise 4.9.4: Good, indifferent, and bad weather

(a) Supply probabilities missing from the wording.

Use good, indifferent, bad. Each column sums to one, so the remaining indifferent-to-bad probability is .3 and bad-to-bad is .3. The matrix is

$$P = \begin{pmatrix} .4 & .5 & .3 \\ .3 & .2 & .4 \\ .3 & .3 & .3 \end{pmatrix}.$$

(b) Tomorrow's bad-weather probability.

Starting from $(.5, .5, 0)$, its bad component after one transition is $.3(.5) + .3(.5) + .3(0) = .3$, or 30%.

(c) Wednesday is two transitions from Monday.

From Monday $(0, .6, .4)$, Tuesday is $(.42, .28, .30)$. Wednesday's good component is $.4(.42) + .5(.28) + .3(.30) = .398$, so the answer is 39.8%.

Exercise 4.9.5: Normalize a two-state balance equation

For $P = \left(\begin{smallmatrix} .1 & .5 \\ .9 & .5 \end{smallmatrix}\right)$, $Pq = q$ reduces to $.9q_1 = .5q_2$, or $q_2 = (9/5)q_1$. Impose $q_1 + q_2 = 1$ to get $(14/5)q_1 = 1$.

Conclusion. $q = (5/14, 9/14)$. Both entries are nonnegative and sum to one.

Exercise 4.9.6: A second two-state equilibrium

Here the balance equation is $.6q_1 = .8q_2$, so $3q_1 = 4q_2$. Together with normalization this gives

$$q = (4/7, 3/7).$$

Substitution gives $Pq = q$, confirming the calculation.

Exercise 4.9.7: Three-state balance with two useful equations

From the first row, $.3q_1 = .1q_2 + .1q_3 = .1(1 - q_1)$, so $.4q_1 = .1$ and $q_1 = 1/4$. From the second, $.2q_2 = .2q_1 + .2q_3 = .2(1 - q_2)$, so $q_2 = 1/2$. Normalization then gives $q_3 = 1/4$.

Conclusion. $q = (1/4, 1/2, 1/4)$. The remaining row equation is satisfied as well.

Exercise 4.9.8: A steady state despite zero transition entries

The second and third row equations give $.5q_2 = .1q_3$ and $.9q_3 = .6q_1$. Thus $q_2 = q_3/5$ and $q_3 = 2q_1/3$. Normalizing gives

$$q_1(1 + 2/15 + 2/3) = 1, \qquad q = (5/9, 2/27, 10/27).$$

Substituting these fractions into the first row gives $5/9$; their sum is one, so they form the required probability vector.

Exercise 4.9.9: One zero can disappear under squaring

The given matrix is stochastic, since it has nonnegative entries and unit column sums. Its square is

$$P^2 = \begin{pmatrix} .2 & 1 \\ .8 & 0 \end{pmatrix}^2 = \begin{pmatrix} .84 & .2 \\ .16 & .8 \end{pmatrix}.$$

Every entry of this power is strictly positive. Therefore P is regular, even though P itself has a zero entry.

Exercise 4.9.10: An absorbing state prevents regularity

For $P = \left(\begin{smallmatrix} 1 & .3 \\ 0 & .7 \end{smallmatrix}\right)$, the first column is $(1, 0)$, and $P(1, 0) = (1, 0)$. Inductively the first column of every positive power remains $(1, 0)$. Thus each power has a zero entry, so P is not regular.

Exercise 4.9.11: Long-run radio preferences

For the matrix in Exercise 4.9.1, the balance equation is $.3q_N = .6q_M$. Thus $q_N = 2q_M$, and normalization gives (a) $q = (2/3, 1/3)$. All entries of P are positive, so it is regular. Theorem 18 implies (b) that the long-run fraction listening to the news is $2/3$, regardless of the initial audience. The finite two-break answer need not equal this limit.

Exercise 4.9.12: No long-run preference between symmetric foods

The matrix of Exercise 4.9.2 sends $(1/3, 1/3, 1/3)$ to itself; every row also sums to one. All entries are positive, so Theorem 18 makes this the unique limiting probability vector. Hence none of the three foods is preferred in the long run: each is chosen with probability $1/3$.

Exercise 4.9.13: The long-run illness rate

For the healthy/ill matrix, $.05q_H = .45q_I$ implies $q_H = 9q_I$. Normalization gives (a) $q = (.9, .1)$. Since this matrix has all positive entries, it is regular; thus (b) the long-run probability that a specific student is ill is .1, or 10%. Theorem 18 applies to every initial probability vector, including certainty of being healthy or ill today, so today's state does not change that limit.

Exercise 4.9.14: Long-run good weather

The third row of the weather matrix gives $q_B = .3$ immediately because $q_G + q_I + q_B = 1$. Then $q_G + q_I = .7$ and $q_I = .3q_G + .2q_I + .4(.3)$. Substitute $q_I = .7 - q_G$ to obtain $.8(.7 - q_G) = .3q_G + .12$, whence $q_G = .4$ and $q_I = .3$. The matrix is strictly positive, so the long-run good-weather probability is 40%.

Exercise 4.9.15: A migration percentage in the long run

With states California and rest of the United States, use

$$P = \begin{pmatrix} .9821 & .0029 \\ .0179 & .9971 \end{pmatrix}.$$

The steady-state equation gives $.0179q_{CA} = .0029q_{rest}$. Since $q_{rest} = 1 - q_{CA}$,

$$q_{CA} = \frac{.0029}{.0179 + .0029} = \frac{29}{208} \approx .13942308.$$

All entries are positive, hence the matrix is regular. The eventual share is approximately 13.9423%. The initial 11.7% affects the transient distributions, but not this limit, under the stipulated constant migration model.

Exercise 4.9.16: Allocate a car fleet across three locations

Order states as City Airport, Downtown, Metro Airport, using the printed *columns* as rental locations and rows as return locations:

$$P = \begin{pmatrix} .90 & .01 & .09 \\ .01 & .90 & .01 \\ .09 & .09 & .90 \end{pmatrix}.$$

The downtown balance gives $.10q_D = .01q_C + .01q_M = .01(1 - q_D)$, so $.11q_D = .01$ and $q_D = 1/11$. The full steady state is $(91/209, 1/11, 9/19)$; its entries sum to one and satisfy $Pq = q$. Because the matrix is positive, it is regular and this steady distribution describes a typical long-run day under the model. For about 2000 cars, downtown gets $2000/11 \approx 181.82$ cars, or approximately **182 cars** rented or ready to rent there.

Exercise 4.9.17: A stochastic matrix has eigenvalue one

(a) Every column of P sums to one, while every column of I sums to one. Thus every column of $P - I$ sums to zero. Adding all other rows to the bottom row produces that column-sum row, which is zero.

(b) The sum of all original rows of $P - I$ is zero. The coefficients of this relation are all 1, not all zero, so the rows are dependent.

(c) A dependent list of n rows spans a space of dimension less than n. Equivalently $\operatorname{rank}(P - I) < n$.

(d) By rank–nullity, $\dim \operatorname{Nul}(P - I) = n - \operatorname{rank}(P - I) > 0$. Thus there is a nonzero x satisfying $(P - I)x = 0$, or $Px = x$.

This proves existence of a nonzero fixed vector; by itself it does not prove its entries can be chosen nonnegative and normalized, which is the additional point mentioned in the exercise.

Exercise 4.9.18: An explicit steady state for every two-state matrix

For $P = \left(\begin{smallmatrix} 1-\alpha & \beta \\ \alpha & 1-\beta \end{smallmatrix}\right)$, the equation $Pq = q$ is $\alpha q_1 = \beta q_2$. If $\alpha + \beta > 0$, choose

$$q = \left(\frac{\beta}{\alpha + \beta}, \frac{\alpha}{\alpha + \beta}\right).$$

Both entries are nonnegative and sum to one, and the balance equation holds, so this is a steady-state vector. It is the unique normalized solution because the balance and sum equations determine both entries. If $\alpha = \beta = 0$, then $P = I_2$. Every vector $(t, 1 - t)$ with $0 \le t \le 1$ is a steady state; in particular $(1, 0), (0, 1)$ are two independent ones. Thus the parenthetical claim means two independent steady-state directions, not only two probability vectors in this case.

Exercise 4.9.19: A row of ones records total probability

(a) Interpret the scalar Sx.

For $S = [\,1 \ \cdots \ 1\,]$, $Sx = \sum_i x_i$. Therefore x is a probability vector exactly when its entries are nonnegative and $Sx = 1$.

(b) Interpret the entries of SP.

Entry j is the sum of column j of P, which equals 1. Hence $SP = S$.

(c) Preserve nonnegativity and the total separately.

Every entry of Px is a sum of products of nonnegative entries, so it is nonnegative. Also $S(Px) = (SP)x = Sx = 1$. By part (a), Px is a probability vector.

Exercise 4.9.20: The square is stochastic

Every column p_j of P is a probability vector. Column j of P^2 is Pp_j, which is a probability vector by Exercise 4.9.19(c). Thus each column of P^2 has nonnegative entries and sums to one, which is exactly the definition of a stochastic matrix.

Exercise 4.9.21: Observe convergence of all columns of powers

(a) Four powers of the supplied positive 4×4 matrix.

Keep full internal precision in the products and round only the displayed entries to four decimal places. The results are

$$P^2 = \begin{pmatrix} 0.2779 & 0.2780 & 0.2803 & 0.2941 \\ 0.3368 & 0.3355 & 0.3357 & 0.3335 \\ 0.1847 & 0.1861 & 0.1833 & 0.1697 \\ 0.2005 & 0.2004 & 0.2007 & 0.2027 \end{pmatrix}.$$

$$P^3 = \begin{pmatrix} 0.2817 & 0.2817 & 0.2817 & 0.2814 \\ 0.3356 & 0.3356 & 0.3355 & 0.3352 \\ 0.1817 & 0.1817 & 0.1819 & 0.1825 \\ 0.2010 & 0.2010 & 0.2010 & 0.2009 \end{pmatrix}.$$

$$P^4 = \begin{pmatrix} 0.2816 & 0.2816 & 0.2816 & 0.2816 \\ 0.3355 & 0.3355 & 0.3355 & 0.3355 \\ 0.1819 & 0.1819 & 0.1819 & 0.1819 \\ 0.2009 & 0.2009 & 0.2009 & 0.2009 \end{pmatrix}.$$

$$P^5 = \begin{pmatrix} 0.2816 & 0.2816 & 0.2816 & 0.2816 \\ 0.3355 & 0.3355 & 0.3355 & 0.3355 \\ 0.1819 & 0.1819 & 0.1819 & 0.1819 \\ 0.2009 & 0.2009 & 0.2009 & 0.2009 \end{pmatrix}.$$

Solving $(P - I)q = 0$ together with $q_1 + q_2 + q_3 + q_4 = 1$ gives

$$q \approx \begin{pmatrix} 0.2816326912 \\ 0.3355052569 \\ 0.181921411 \\ 0.2009406409 \end{pmatrix}.$$

The columns are rapidly becoming nearly equal to this fixed probability vector.

(b) Powers of the more slowly mixing 3×3 matrix.

$$Q^{10} = \begin{pmatrix} 0.8222 & 0.4044 & 0.5385 \\ 0.0324 & 0.3966 & 0.1666 \\ 0.1453 & 0.1990 & 0.2949 \end{pmatrix}.$$

$$Q^{20} = \begin{pmatrix} 0.7674 & 0.6000 & 0.6690 \\ 0.0637 & 0.2036 & 0.1326 \\ 0.1688 & 0.1964 & 0.1984 \end{pmatrix}.$$

$$Q^{30} = \begin{pmatrix} 0.7477 & 0.6815 & 0.7105 \\ 0.0783 & 0.1329 & 0.1074 \\ 0.1740 & 0.1856 & 0.1821 \end{pmatrix}.$$

$$Q^{40} = \begin{pmatrix} 0.7401 & 0.7140 & 0.7257 \\ 0.0843 & 0.1057 & 0.0960 \\ 0.1756 & 0.1802 & 0.1783 \end{pmatrix}.$$

$$Q^{50} = \begin{pmatrix} 0.7372 & 0.7269 & 0.7315 \\ 0.0867 & 0.0951 & 0.0913 \\ 0.1761 & 0.1780 & 0.1772 \end{pmatrix}.$$

$$Q^{60} = \begin{pmatrix} 0.7360 & 0.7320 & 0.7338 \\ 0.0876 & 0.0909 & 0.0894 \\ 0.1763 & 0.1771 & 0.1767 \end{pmatrix}.$$

$$Q^{70} = \begin{pmatrix} 0.7356 & 0.7340 & 0.7347 \\ 0.0880 & 0.0893 & 0.0887 \\ 0.1764 & 0.1767 & 0.1766 \end{pmatrix}.$$

$$Q^{80} = \begin{pmatrix} 0.7354 & 0.7348 & 0.7351 \\ 0.0881 & 0.0887 & 0.0884 \\ 0.1764 & 0.1766 & 0.1765 \end{pmatrix}.$$

Its second equilibrium equation gives $q_2 = q_3/2$, and its first gives $.03q_1 = .05q_2 + .10q_3$. Normalization then yields

$$q = (25/34, 3/34, 3/17) \approx (.73529412, .08823529, .17647059).$$

The columns approach this vector more slowly, so an arbitrary fixed power such as 100 is not a universal certificate of four-place accuracy.

(c) Explain the observation with Theorem 18.

P is positive, hence regular. Q^2 has all entries positive (the only zero in Q acquires a positive two-step route), so Q is regular too. Column j of P^k is $P^k e_j$, the evolving probability vector from the initial state e_j. Theorem 18 says each such vector tends to the same unique steady state. The same reasoning applies to Q^k and explains the observed column limits for any regular stochastic matrix.

Exercise 4.9.22: A reproducible comparison of two numerical methods

What we must show. Run both methods, report measured times, and distinguish numerical approximation from a mathematical convergence guarantee. A matrix program such as NumPy has no fixed largest permitted order; the practical limit depends on memory and desired runtime. The

following experiment uses positive random matrices through order 1024, the largest order tested here, rather than claiming an intrinsic software size limit. Normalize each random positive column to obtain a regular stochastic matrix. For method 1 solve $(P - I)q = 0$ with its final equation replaced by $\sum_i q_i = 1$. For method 2 use column one of P^{100}.

The following table records actual execution times in seconds on this workspace, with one BLAS thread; input generation is outside the timing. Each method is invoked by one matrix-program call after the shared setup. Human typing time is user dependent and is not fabricated as an execution measurement.

Order	Direct solve	Power 100	Maximum difference
64	0.000149	0.000120	$6.09e-16$
128	0.000173	0.000829	$3.93e-16$
256	0.000788	0.005619	$5.89e-16$
512	0.004078	0.043638	$7.01e-16$
1024	0.028829	0.329893	$7.84e-16$

Here is the complete experimental core; start Python with `OPENBLAS_NUM_THREADS=1` for the stated thread setting.

```
import numpy as np
from time import perf_counter
rng = np.random.default_rng(4922)
for n in [64, 128, 256, 512, 1024]:
    P = rng.uniform(.1, 1, (n, n)); P /= P.sum(axis=0)
    A = P - np.eye(n); A[-1, :] = 1
    b = np.zeros(n); b[-1] = 1
    start = perf_counter()
    q = np.linalg.solve(A, b)
    direct = perf_counter() - start
    start = perf_counter()
    v = np.linalg.matrix_power(P, 100)[:, 0]
    power = perf_counter() - start
    print(n, direct, power, np.max(abs(q-v)))
```

For these well-mixed positive samples, the two vectors agree to less than 8×10^{-16} in maximum coordinate error. This empirical accuracy is not guaranteed for every regular stochastic matrix: a slowly mixing matrix can require far more than 100 steps, as Exercise 4.9.21 illustrates. Solving the normalized linear system avoids choosing a mixing time and was faster for the larger tested orders. Computing a power is easy to request and can reveal transients, but a large power performs more matrix products and still needs a convergence check. For finding only the steady state, the direct normalized solve is preferable in this experiment.